

Dinesh Kumar Rajakumaran

✉ dinulkar2000@gmail.com — 📞 +43 677 61009477

🌐 Dinesh Kumar Rajakumaran — 🔄 Dinulkar

📍 Wels, Austria



Profile

System Engineer with 3+ years of experience in the automotive industry and a Master's degree in Automotive Mechatronics and Management. Experienced in vehicle testing, system validation, simulation, and control engineering. Strong knowledge of MATLAB/Simulink, vehicle dynamics, powertrain systems, and model-based development. Skilled in data analysis, performance evaluation, and cross-functional collaboration throughout the product development process. Passionate about developing efficient and reliable automotive systems for next-generation mobility solutions.

Skills

- **System Engineering:** System Modeling, Requirements Analysis, Verification & Validation, Functional Decomposition, System Integration, Engineering Documentation
- **Powertrain & Transmission Systems:** Transmission Systems, Transmission Control Systems, Hybrid Powertrains, Vehicle Dynamics, Energy Management, Drive Systems & E-Mobility
- **Simulation & Modeling:** MATLAB, Simulink, Model-Based Development, Dynamic System Modeling, State-Space Modeling, PID Control, Model Predictive Control (MPC)
- **Testing & Validation:** Vehicle Testing, Functional Validation, Performance Evaluation, Root Cause Analysis, Simulation-Based Testing, Data Acquisition and Analysis
- **Data Analysis:** MATLAB Data Processing, Signal Analysis, Performance Assessment, Engineering Data Visualization, Statistical Analysis
- **Automotive Standards & Tools:** ISO 26262, AUTOSAR Fundamentals, CAN Communication, CANoe, Automotive Ethernet Fundamentals
- **Professional Skills:** Analytical Problem Solving, Cross-Functional Collaboration, Technical Reporting, Project Coordination, Stakeholder Communication, Structured Engineering Approach

Professional Experience

BRP-Rotax GmbH & Co KG, Austria

Oct 2024 – Jun 2025

FH Intern – Control Systems Engineering

- Developed high-fidelity nonlinear vehicle and powertrain-related dynamic models using MATLAB/Simulink for performance evaluation and control strategy development.
- Performed system-level analysis of vehicle behavior across varying operating conditions to evaluate stability, efficiency, and performance objectives.
- Designed and validated control strategies using simulation-based approaches and correlated results with expected operational requirements.

- Conducted parameter studies and trade-off analyses to support engineering decisions related to vehicle dynamics and system performance.
- Collaborated with multidisciplinary engineering teams to align system requirements, validation activities, and development objectives.
- Generated technical documentation, validation reports, and engineering recommendations to support product development activities.

Continental AG, Austria

Feb 2024 – Jun 2024

Master's Project Student

- Developed sensor data evaluation workflows using MATLAB ROS Toolbox and Foxglove Studio.
- Collaborated with engineering teams to support validation activities and technical issue resolution.
- Configured communication interfaces for reliable real-time data acquisition and analysis.
- Prepared technical reports and presented validation findings to project stakeholders.

Royal Enfield Global Headquarters, India

Mar 2022 – Sept 2023

Vehicle Testing Engineer

- Executed vehicle testing and validation activities for ABS and braking systems during product development.
- Coordinated testing schedules, data collection, and performance evaluations across multiple development programs.
- Collaborated with Tier-1 suppliers including Bosch and Continental to resolve technical issues and support validation activities.
- Contributed to project reporting, process improvements, and successful completion of regulatory testing requirements.

Education

University of Applied Sciences Upper Austria

2023 – 2025

M.Sc. Automotive Mechatronics and Management

Relevant Coursework: Transmission Systems, Transmission Control Systems, Hybrid Powertrain Control, Vehicle Dynamics, Drive Systems and E-Mobility, Model-Based Engineering, MATLAB/Simulink, Functional Safety (ISO 26262), Mechatronic Systems Validation, System Integration, Requirements Engineering, Analysis, Calibration, Agile Methodologies, Project Management, Program Management

Rajalakshmi Engineering College

2017 – 2021

B.E. Automobile Engineering

Relevant Coursework: Vehicle Dynamics, Automotive Electronics, Embedded Systems, Thermodynamics, Fluid Mechanics, Manufacturing Technology

Projects

Hydrofoil Vehicle Control and Validation (Master's Thesis)

- Developed nonlinear dynamic models and state-space representations for system-level performance evaluation.

- Designed and validated PID and Model Predictive Controllers using MATLAB/Simulink.
- Performed simulation-based analysis of vehicle performance under varying operating conditions and actuator constraints.
- Conducted trade-off studies to balance stability, control performance, and system limitations.

Sensor Data Evaluation using ROS and MATLAB

- Developed a data acquisition and analysis workflow using MATLAB ROS Toolbox and Foxglove Studio for automotive sensor evaluation.
- Processed and analyzed large-scale radar sensor datasets to support system validation and performance assessment activities.
- Performed data-driven investigations to identify measurement inconsistencies, validate sensor behavior, and support engineering decision-making.
- Developed visualization and reporting tools for efficient interpretation of test results and communication of technical findings.
- Collaborated with engineering teams to verify sensor performance and support troubleshooting during development and validation phases.

Languages

Tamil (Native), English (C1), German (B1)